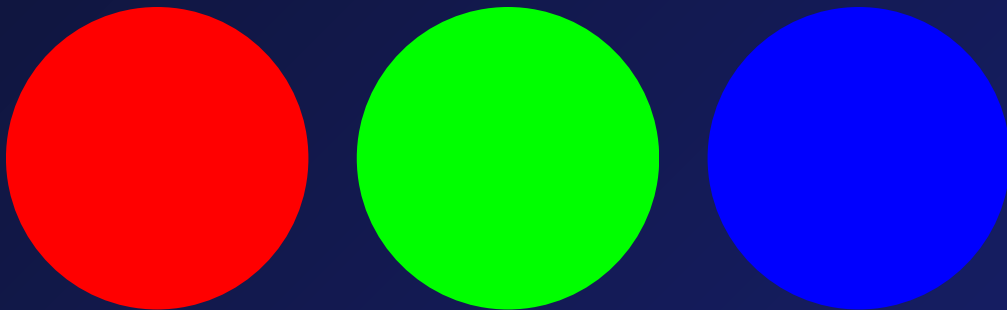


Colour Mixing Light Show

Session 1: Discover the Magic of RGB!



PLANRS ACADEMY MASTER COURSE

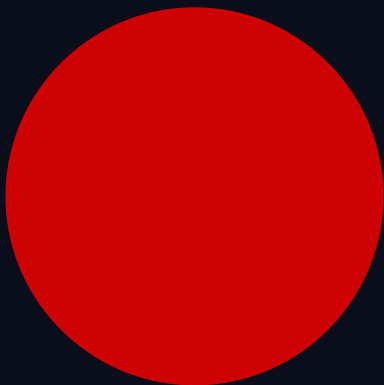
The Secret of White Light

What if I told you that the colour **WHITE** is actually made of all the colours mixed together? [cite: 813]

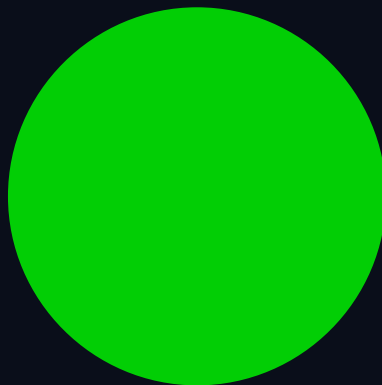
You have seen beautiful lights during **Diwali** or colourful spotlights on your school stage. Today, we find out the secret behind those colours! [cite: 815]

The Magic Trio

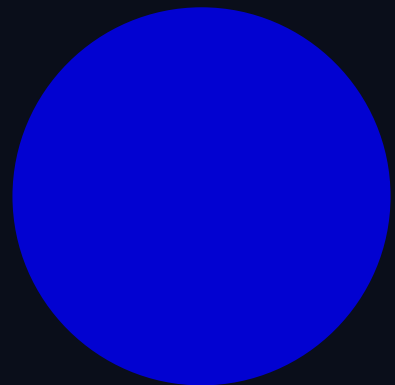
In the world of light, everything starts with three "Magic" primary colours. You cannot make these by mixing anything else! [cite: 801, 802]



RED



GREEN



BLUE

Call them out together: RED! GREEN! BLUE! [cite: 819]

Where is the Magic?

We see these magic colours all around us in India! [cite: 803]

Diwali Decorations

Tiny LED bulbs on homes in Delhi or Mumbai glow in red, green, or blue light independently to make the city sparkle! [cite: 803, 804]

Traffic Signals

From Bangalore to Kolkata, every traffic light uses big, bright Red and Green lights to tell cars when to stop and go! [cite: 866]

Your Screen

Look at a phone or TV screen. It actually uses millions of tiny Red, Green, and Blue dots to show you all your favorite cartoons! [cite: 867]

The Colour Mixing Show

What happens when two primary colours of light meet? It's like a magic trick!
[cite: 821, 824]



YELLOW! (Like stage spotlights during school functions) [cite: 805, 823]



MAGENTA! (A bright pinkish-purple) [cite: 805, 824]

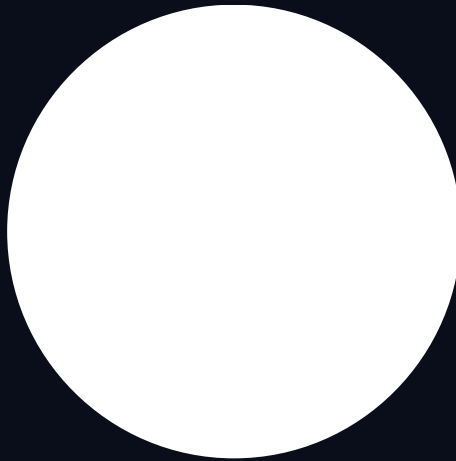


CYAN! (A bright blue-green) [cite: 805, 824]

Guess First! What do you think happens when we mix ALL THREE? [cite: 852]

The Big Reveal: White Light

When **Red**, **Green**, and **Blue** light all come together, they create pure **White Light**! [cite: 806, 825]



Indian Example: The white LED bulbs and tube lights in your home or classroom are actually mixing these colours right now! [cite: 806, 868]

What about Darkness? When there is NO light, we see **Black**. Think of a power cut at night—it goes dark because there is zero light! [cite: 808, 809]

| Light is Different from Paint!

Don't be confused! Mixing light is a bit like magic, while mixing paint is like cooking. [cite: 812, 828]

Mixing Paint (Like Camlin sets!)

When you mix paints, the colour gets **DARKER**. [cite: 810]

Example: Blue + Yellow paint = Green. [cite: 810, 812]

Mixing Light (The Magic Way!)

When you mix light, the colour gets **BRIGHTER**! [cite: 811, 829]

Example: Red + Green light = Yellow. [cite: 811]

Remember: Light adds more brightness! [cite: 829]

You are a Colour Scientist!

Let's Recap! [cite: 834]

- Primary Colours of Light: **Red**, **Green**, **Blue**. [cite: 895]
- Red + Green = **Yellow**. [cite: 897]
- All three together = **White Light**. [cite: 834]

Success Criteria [cite: 895-899]

- ✓ I can name the 3 magic colours of light. [cite: 895]
- ✓ I can predict what happens when lights mix. [cite: 896]
- ✓ I can name one place in India where coloured lights glow. [cite: 899]
- ✓ I participated in the light mixing activity! [cite: 900]

 **CONGRATULATIONS!** 

You have finished Session 1 of the Light Show!